

## Haiyang Zhang

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### Professional Summary

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Senior Frontend Engineer with 8 years of experience building React, TypeScript, and React Native applications for enterprise platforms, healthcare products, and cloud-deployed full-stack web systems. Strong in front-end architecture, product-minded execution, API integration, performance optimization, and hands-on backend development with Python, FastAPI, PostgreSQL, authentication flows, and Google Cloud-based deployment.

### Core Skills

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- Languages: HTML, CSS, JavaScript, TypeScript
- Frontend & Mobile: Vue.js, React, Next.js, React Native, Expo, Responsive Web Applications, Cross-Platform iOS/Android Development
- State Management: Vuex, Pinia, Zustand
- Testing & Quality: Jest, Cypress, Playwright, React Testing Library, End-to-End Testing, UI Testing
- Tooling: Webpack, Vite, Docker Compose, Nginx
- DevOps & Cloud: Docker, Kubernetes, CI/CD, Google Cloud Platform (GCP), AWS, Azure
- Architecture & UX: Front-End Architecture, Reusable Components, Component Design, Design Systems, Workflow Systems, Performance Optimization, Accessibility
- Backend & Integration: Node.js, Express.js, Python, FastAPI, REST API Integration, GraphQL, PostgreSQL, SQLAlchemy, Prisma, Auth.js
- Delivery: CI/CD, App Store, Google Play
- Healthcare & Security: FHIR, EHR Integration, Healthcare Data Processing, Data Privacy, Application Security
- Additional: ethers.js, Blockchain Application Development

### Professional Experience

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#### Novo Vivo

Frontend Engineer / Software Engineer Sep 2018 - Present

##### Privacy Computing Platform Modernization

2018 - Present

- Led front-end development for a large-scale privacy computing platform spanning model management, inference testing, online and offline workflows, access control, service monitoring, and operational dashboards across multiple interconnected subsystems.
- Evolved an early project-based implementation into a standardized platform by abstracting shared capabilities across data resources, federated modeling, privacy-preserving query, approval workflows, and system management, improving reuse across follow-on client projects.
- Built interfaces for model listing, inference configuration, execution history, task orchestration, and result inspection, improving workflow usability and reducing manual steps for enterprise users.
- Partnered with backend and algorithm teams on REST and GraphQL API contracts, plus asynchronous task flows for model deployment, runtime tracking, and result delivery.
- Reduced duplicated front-end implementation by an estimated 30% to 40% through reusable business components, shared interaction patterns, and design-system standards, accelerating delivery across similar modules.
- Optimized workflow-canvas and core management screens, improving first-screen rendering and interaction responsiveness by an estimated 30%+ in drag-and-drop scenarios.
- Strengthened delivery quality with Cypress and Playwright regression coverage for complex workflows while improving CI/CD release confidence and mentoring up to 4 engineers.

#### Lavita Health App

2023 - Present

- Supported 100,000+ downloads across iOS and Android by developing a cross-platform healthcare application with Expo and React Native that delivered AI-driven health guidance in a privacy-focused product environment.
- Built mobile ingestion flows for personal health records, enabling users to import data from major EHR systems including Epic and Cerner.
- Implemented FHIR-based transformation and normalization logic, converting standards-compliant medical records into product-ready data for downstream AI workflows.
- Added fallback handling across the import pipeline so failures in one step would not block later API requests or downstream data retrieval.
- Improved cross-platform quality and release readiness with Jest coverage for critical user flows and application logic across iOS and Android.
- Managed App Store and Google Play release workflows independently, increasing release ownership and reducing cross-functional handoff delays.
- Integrated Google NotebookLM to generate healthcare blog and podcast content, and built Web3 token flows on Theta's subchain including registration, balance retrieval, and token transfer.

- Supported privacy-first health-data flows using a hybrid encryption approach, combining asymmetric key exchange with symmetric encryption for larger payloads before downstream processing and secure API integration.

## **Birding Copilot**

**Creator / Full-Stack Engineer (Personal Project) 2026 - Present**

- Built a bird-finding web application with a split Next.js and FastAPI architecture that turns structured searches into recommendation-first result pages backed by real eBird data, reducing the effort needed to compare hotspots and recent sightings manually.
- Designed the backend query pipeline with FastAPI, SQLAlchemy, and PostgreSQL for query normalization, evidence collection, report generation, database persistence, and structured payload storage.
- Integrated server-side eBird lookups for species, region, hotspot, and recent/notable observations while ranking candidate sightings with recency-first logic.
- Implemented the web layer with Next.js App Router, TypeScript, Prisma, and Auth.js, including server-side API routes, authentication flows, anonymous search support, and saved report retrieval.
- Added Google Analytics to track search behavior and result-page engagement, creating a feedback loop for improving query flow, recommendation UX, and future product decisions.
- Deployed the application on Google Cloud Platform (GCP) with Docker Compose and Nginx reverse proxy, establishing a clean production setup with separate `web`, `api`, and database runtime boundaries.

## **Education**

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### **Syracuse University**

**M.S.E. in Computer and Information Science, 2018**

### **Beijing University of Technology**

**B.S.E. in Internet of Things, 2016**